

Interview Summary

Interview goal

Our goal was to understand what makes help useful or unhelpful for STEM students when they are trying to learn or solve problems. We deliberately framed this around help in general rather than AI, hoping that the way students seek and judge help from other sources could show us how AI support might become more useful for learning.

Participants

Our target population is STEM students. We used **convenience sampling** and interviewed 12 friends and colleagues studying STEM subjects, mostly at BSc level, with one Master's student and one recent graduate. We intentionally avoided interviewing only Computer Science students, so that our results would be more overarching and the interviews would not be tailored to what we expected Computer Science students to say. The final sample covers Medicine, Pharmacy, Mathematics, Mechanical Engineering, Electrical Engineering, Robotics, Industrial Engineering, Computational Biology, Computer Science, Information Systems and Quantitative Finance.

Main questions

The full script is in the supplementary materials. The questions followed four steps:

- **Profiling:** field of study, course difficulty, typical problems and how quickly they seek help. This gave us enough background to differentiate participants and look for patterns across fields.
- **A concrete example:** the last time they got really stuck and what happened until they moved on. We asked for a raw account first, so as not to prime them on whether the help they received was useful.
- **Useful versus unhelpful help:** what they usually do when stuck, how they decide to ask, examples of useful and unhelpful help, and whether they could later solve similar problems on their own. None of the questions mentioned AI, to get a general view of how students seek help, although from experience with our peers we expected AI to come up.
- **Reflection:** what kind of help still lets them feel they are solving the problem themselves, and how they view the timing of asking for help. We placed these last, expecting participants to have more to say after the conversation.

Key takeaways

- **AI is the default source of help.** Every participant brought up AI on their own, even though no question mentioned it.
- **Overly complex help backfires.** Long, elaborate explanations, or help that assumed knowledge participants did not have yet, led to frustration or was simply not useful.
- **Step-by-step help feels more natural.** Participants preferred help that comes in small steps and leaves part of the work to them, rather than the complete solution at once.
- **Sometimes pointing the way is enough.** Some participants said help was most useful when it did not even give the steps, but pointed them to where in their course material to look.